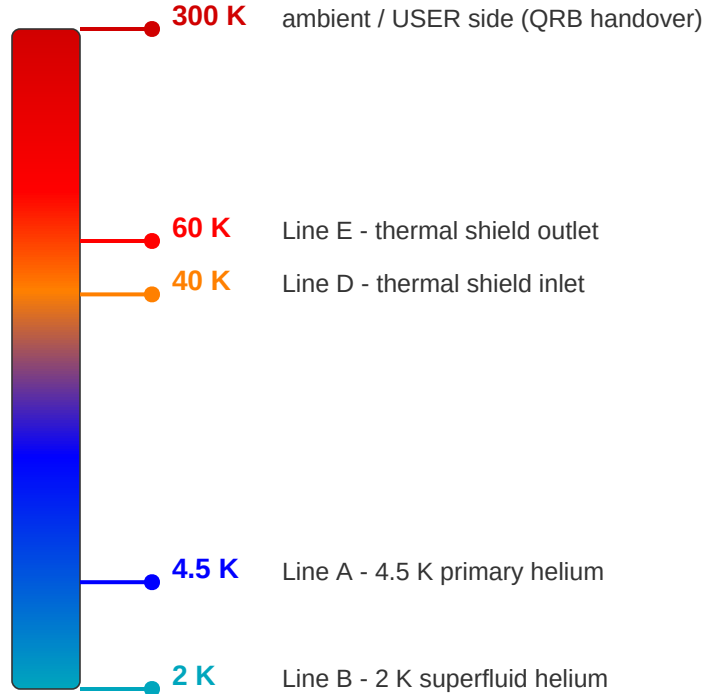


Cryogenic temperature ladder of the distribution lines and the Line W warm-return gradient

TEMPERATURE LADDER



Why Line W has a gradient

Line W returns helium from the cold cryomodule back to the warm piping system (WPS). It must be re-warmed from cryogenic to ambient before reaching the user-side QRB handover.

- Cold end : 4.5 K, leaving the 4.5 K circuit / cold box
- Mid-run : electrical trim heater + ambient heat in-leak
- Warm end : ~300 K (ambient) at the QRB handover (NA.CP03)

Design data: ~2.5 g/s · 6 bar · DN20 · SS304

The same gradient strip appears on every production sheet (layer 15_Temperature_Gradient) and in the MAIN-LINES schematic.

LINE W - WPS WARM RETURN GRADIENT (~2.5 g/s - DN20 - SS304)

